

Homework 1 — Notebook and AI

Quantitative Reasoning II (INF 1220)

Due Monday, September 21, 11:59pm, on Populi

Work through the notebook and hold one study conversation with an LLM. Turn in both to Populi by the due date. Read *Using AI for Study and Data Analysis*, on the [Materials](#) page, before you start.

1. Setting up

You need a Google account. There is nothing to install.

1. Open the notebook: <https://colab.research.google.com/drive/1Vq-05bLU2rieu-YT67TghfZgB0cInL8M>
2. Click **File** → **Save a copy in Drive**. Do this before running or typing anything. The copy shared with you is read-only, so without this step nothing you do will save.
3. Its name will start with “Copy of”. Change this to include your last name.
4. Work in your copy. Any work you do in the original will not be saved.
5. Run the cells in order. **Shift** + **Enter** runs the cell you are in; **Runtime** → **Run all** runs everything.

Gemini is built into Colab, and a failed cell gets an **Explain error** button. Hold the study conversation separately, at gemini.google.com, since that is where you can get a link to turn in.

2. Python Notebook

Run the code, answer the questions, and do the tasks the notebook sets. Answers go in the notebook itself, in the cells marked for them.

The notebook contains errors. Finding and correcting errors is part of the assignment. Do not assume anything in the notebook is correct and do not assume a result is correct because the code runs without errors.

3. LLM Study

Hold one conversation with an LLM covering these two topics:

1. Descriptive statistics and skew
2. Conditional probabilities and base rates

Follow the expectations in *Using AI for Study and Data Analysis*. Finish with the prompt from *Evaluate the conversation*, pasted as written, with those two topics as the list.

4. Assignment Submission

Turn in two items to Populi:

1. Your notebook, fully run. In Colab: **Share** → **Anyone with the link** → **Commenter**, then submit the link to the Populi assignment.
2. Your study conversation, shared the same way.

Points will be taken off if a I try to open a link and I cannot see your work. Check both of your links in a private browsing window before you submit.

Sharing an LLM conversation shares everything in the thread, so do not put anything private in the conversation you turn in.

5. If you do not use Colab or Gemini

You are not required to use Colab or Gemini (this is just the quickest set of tools to get spun up). If you choose to use another set of tools, (1) you are responsible for their functionality, and (2) upload your notebook to your Google Drive, change the sharing settings accordingly, and submit the link to the Populi assignment. Same goes for your LLM study conversation.